

BRAC IT CodeSprint 2026

Project BRD

1. **Project Name** Probashi Shield

2. **Project Owner** Umme Jannath Tamanna

3. Problem Statement

- People go abroad through middlemen (brokers), but the person migrating is often unsure whether the middleman is authorized and authentic.
- There is no reliable way to verify if the job offer is genuine and if the associated information is accurate, leaving migrants highly vulnerable to fraud.
- Once the money is paid and fraud occurs, there are virtually no mechanisms or legal processes to recover the lost funds.

4. Proposed Solution

Probashi Shield lets a worker — or a family member helping them decide — photograph the document they've been given (a job offer, visa page, contract, or agency business card) and receive an instant, plain-Bangla verdict before any money changes hands. The flow has five steps:

- Capture — the user photographs the document with their phone camera; no scanner needed.
- Extract — OCR reads the image and pulls out the agency name, BMET license (RL) number, destination country, salary, fees, and job title.
- Verify — the extracted license number is cross-checked against BMET's public licensed-agency registry to confirm the agency is currently licensed, and that the registered name matches the name on the document (to catch a forged document reusing someone else's real license number).
- Flag — a rules engine checks for clear-cut violations (fee above the legal ceiling, malformed or missing license number), while an AI model reads the extracted Bangla text for softer red flags — unrealistic salary, vague terms, pressuring language, mismatched details.
- Report — the user sees one of three verdicts — Verified / Caution / High Risk — with the specific reasons explained in plain Bangla and a suggested next step (proceed, ask the agency to clarify, or report to BMET / RAWA).

No account or login is required — this needs to work for a one-time, high-stakes decision with zero friction.

5. Target Users

- Primary: prospective migrant workers who have received a job offer and are deciding whether to pay a recruiting fee.
- Secondary: a family member helping vet the offer on the worker's behalf — this is often a household decision, not an individual one.
- Design constraints that follow: Bangla-first interface, low digital literacy, low-end Android phones, and potentially poor connectivity.

6. Key Features

- Photo-based document capture — photograph an offer letter, visa page, contract, or agency business card directly in the app; no scanner or printer needed, since most workers only have a phone camera.
- Capture guidance — the app prompts the user to avoid glare and frame the full document before submitting, since photo quality directly drives OCR accuracy.
- OCR field extraction with confirmation — pulls the agency name, BMET license (RL) number, destination country, salary, fees, and job title from the photo, and shows the extracted fields back to the user to confirm or correct before verification runs.
- Live BMET registry cross-check — checks the license number against a synced mirror of BMET's public registry and returns its status (Active / Inactive / Expired / Suspended / Cancelled / Not Found), along with a "last updated" timestamp.
- Name-match validation — compares the agency name on the document to the name registered against that license number, catching a forged document that reuses someone else's real, valid license number.
- Rule-based red-flag checks — automatically flags a fee above the government's legal ceiling for that destination country, or a malformed or missing license number.
- AI-based scam-pattern detection — reads the extracted Bangla text for softer red flags that don't reduce to a fixed rule, such as vague job descriptions, inconsistent details, or pressuring language.
- Plain-Bangla explanations, never a bare score — every flag, rule-based or AI-based, comes with a specific reason in plain Bangla, so the user understands exactly what's wrong.
- Single, clear verdict with next steps — combines every check into one of three outcomes (Verified / Caution / High Risk), with a concrete next action: proceed, ask the agency to clarify, or report to BMET / RAWA.
- No account or login required — the full capture-to-verdict flow works with zero setup, since this is a one-time, high-stakes check that needs to happen fast.

7. AI Component

- OCR extraction — a Bangla-capable OCR model reads real-world phone photos (handling skew, glare, and stamps) and converts them into structured fields. Candidates under evaluation include Google Cloud Vision (fastest to integrate) and bbOCR, an open-source pipeline built specifically for Bangla documents captured under distortion and poor lighting.
- Scam-pattern detection — an AI/LLM component reads the extracted Bangla terms and flags softer red flags that don't reduce to a simple threshold — vague job descriptions, inconsistent details, pressuring language — and generates the plain-Bangla explanation shown to the user. This runs alongside, not instead of, deterministic rule checks for clear-cut violations such as fee-ceiling breaches.

8. Expected Outcome / Impact

- Workers get a verification check at the exact moment it can still change their decision — before a fee is paid — instead of after fraud is discovered abroad, when nothing can be recovered.
- Reduces the number of workers defrauded by fake or unlicensed recruiting agents each year.
- Protects household savings that would otherwise be lost to fraud, and protects the reliability of remittance income that is a major pillar of Bangladesh's economy.
- Converts a public but hard-to-reach government resource (BMET's registry and fee ceilings) into something usable at the actual point of decision.

9. Data / Resources Required

- BMET's public licensed-agency registry (oep.gov.bd/agencies) — license number, agency name, address, phone, status, and validity date for roughly 2,857 agencies. No public API exists, so this requires a scheduled scrape/sync job.
- Government fee-ceiling data per destination country, to power the rule-based fee check.
- A Bangla-capable OCR engine — candidates evaluated: Google Cloud Vision, bbOCR (open source), Tesseract.
- An AI/LLM model capable of reading and reasoning over Bangla text for scam-pattern detection.
- Reference salary-range data by job category and destination country, to judge “unrealistic salary” claims.
- Optional secondary cross-check: BAIRA's member-agency list (baira.org.bd).

10. Success Criteria

- A worker can go from photographing a document to seeing a verdict in under one minute.
- License verification produces no false “Verified” results against a test set — including documents that reuse a real license number under the wrong agency name, which must be caught by the name-match check.
- OCR extraction is usable on real, unedited phone photos of Bangla documents, validated against a test set of real offer letters and visas — not clean scans.
- In a pilot (e.g., during a pre-departure briefing session), a majority of workers report that the verdict and explanation were understandable in plain Bangla.
- At least one documented case in the pilot where a worker changed a payment decision based on the tool's verdict.

Why Should You Build This?

Every year, hundreds of thousands of Bangladeshi workers lose money to fake job offers and unlicensed agents. The money these workers send home is one of the biggest parts of Bangladesh's economy. The government already keeps a list of licensed agencies, but workers have no easy way to check it before they pay. Probashi Shield fixes this: take one photo, get a simple answer in plain Bangla, before any money changes hands. Once fraud happens, there is almost no way to get the money back. This tool stops that from happening in the first place, protecting real families from losing their life savings.